Collapse Around the Circuit: Induction Heads Survive Recursive Training While Their Errors Are Inherited

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Abstract

We study model collapse inside a mechanism we can read: the induction circuit of Singh et al. (2024). A two-layer attention-only transformer learns an in-context symbol task and is retrained for up to 20 generations on pools that mix a fraction λ of real data with data sampled from its predecessor. Across 311 runs the circuit survives: query accuracy never falls below 0.979 and ablating layer 1 costs 0.69–0.74 accuracy at every λ and generation. What degrades is the model's free generative choice. The probability it places on symbols absent from the context follows a two-parameter inheritance law, $m_{g+1} = a(1-\lambda)m_g + \varepsilon$ ($a = 0.951$, $R^2 = 0.994$), which predicts held-out real-data fractions. Removing those off-context samples from the synthetic pools cuts the gain a to 0.19, and training-data attribution finds them 12–23$\times$ over-represented among the most influential examples: the errors are inherited through the data, not produced by a damaged circuit. A pre-registered extension of the law to higher sampling temperatures failed four of five tests. Runaway collapse occurred only with no real data at all, so real data protects more strongly than the linear law predicts.

Keywords: model collapse; induction heads; in-context learning

Introduction

Language models adapt to patterns shown only in their prompt, and much of this in-context learning (ICL) is attributed to *induction heads*: a previous-token head copies each token's predecessor into its position, and an induction head one layer later attends from the current token to an earlier match and copies what followed it (Olsson et al., 2022). Singh et al. (2024) isolate this circuit in a minimal symbol-copying task. Separately, models trained recursively on generated data suffer *model collapse*, losing the tails of the data distribution (Shumailov et al., 2023, 2024); collapse is avoided when enough real data remains (Bertrand et al., 2023) or when data is accumulated rather than replaced (Gerstgrasser et al., 2024). These lines of work have not met. We ask whether recursive training damages the ICL circuit itself or only what the circuit produces, and we make a quantitative, pre-registered prediction about when collapse runs away.

Background Attention patterns suggest a circuit but are not causal evidence, so we follow Conmy et al. (2023) and ablate heads. To trace collapse back to specific training examples we use gradient-based data attribution (Pruthi et al., 2020), in the spirit of Chen et al. (2026).

Methods

Task and model. Each sequence gives $K{=}4$ symbols (from 16) a random label (from 6). A study block lists the pairs and a query block repeats the symbols in an independent order; the model predicts each label, which only content-based matching can solve. The model also freely generates one more in-context symbol ($sq2$), which is drawn uniformly and is the degree of freedom that can collapse. The model has 2 attention-only layers of 4 heads ($d{=}64$, rotary positions, tied embeddings, cosine learning-rate decay). Hyper-parameters were chosen on validation loss and confirmed once on a held-out test seed.

Collapse pipeline. Generation 0 trains on a 20k-sequence real pool. Generation g trains a *fresh* model on a 20k pool with a fraction λ of real sequences and the rest sampled from generation $g{-}1$ at temperature $T{=}1$. Prompts stay real and all answers are model-generated (*prompt-real* regime). Each pool is split 90/10 into train and validation; test data is fresh real data. A generation must pass a strict gate (validation accuracy and a causal layer-1 check) or is retrained, which happened 20 times in 680 main-sweep generations with no final failures. We sweep $\lambda \in \{0, 0.1, 0.25, 0.5, 1\}$ with 10 seeds, for 12 generations (20 at $\lambda{=}0$).

Metrics. *Off-context mass*, the probability $p(sq2 \mid \text{context})$ places on symbols not in the context, measures hallucination; query NLL on real data measures calibration. The circuit is measured causally, as the accuracy lost when every layer-1 head is mean-ablated. This is agnostic to how the circuit is implemented, which matters because 5% of generations use a non-canonical variant that attention probes misread. We fit the inheritance law on $\lambda \in \{0, 0.25, 1\}$ and test it on $\{0.1, 0.5\}$. Interventions remove off-context or wrong-label samples from the synthetic pools, accumulate pools across generations, change the pool size, inject a one-off 5% label-noise pulse, and vary depth, width and task size.

Pre-registration. From one calibration seed, the law was fitted per temperature $T \in \{0.7, 1.3, 1.6\}$. It predicts a critical fraction $\lambda^* = 1 - 1/a(T)$ whenever $a(T) > 1$. Five pass/fail tests were committed to git before the confirmatory runs (5 seeds per cell).

Results

The circuit survives; its outputs do not. Query accuracy stays $\geq$0.979 in every generation, and the accuracy lost by ablating layer 1 is 0.71 at generation 0 and 0.69–0.74 at the last generation for every λ (Figure 1c). Calibration erodes instead: without real data, query NLL rises from 0.006 to 0.030 by generation 19 while accuracy is 0.996 (Figure 1b).

An inheritance law. Off-context mass grows from 0.013 to 0.138 by generation 19 without real data, and to 0.057, 0.035, 0.022 and 0.012 by generation 11 at $\lambda = 0.1, 0.25, 0.5$ and 1. The law fits all 630 transitions with $a = 0.951$ (95% CI 0.942–0.959), $\varepsilon = 0.0105$ and $R^2 = 0.994$. Fitted without $\lambda \in \{0.1, 0.5\}$, it predicts them with RMSE 0.003, within one seed SD at 68% of generations (Figure 1a). It overshoots the plateau: for $\lambda{=}0$ it predicts $m^* = 0.21$, while mass is 0.13 at generations 16–19 and still rising. The gain holds across 3- and 4-layer, MLP and wider models ($a = 0.94$–1.04, $R^2 \geq 0.98$).

The errors are inherited through the data. Removing off-context samples from each synthetic pool cuts the gain from 0.95 to 0.19, while removing wrong labels leaves it at 0.95. Attribution agrees: off-context samples are 12–23$\times$

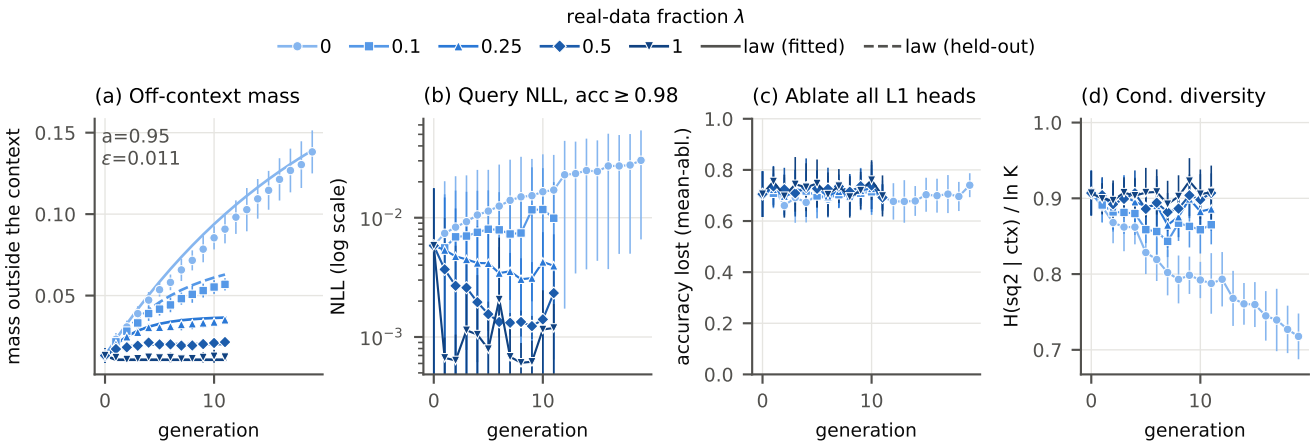


Figure 1: Main sweep (10 seeds, mean $\pm$ sd). (a) Off-context mass with the inheritance law; solid lines are fitted fractions, dashed lines are held-out predictions. (b) Query NLL rises without real data while accuracy stays $\geq$0.98. (c) Accuracy lost when all layer-1 heads are mean-ablated stays near 0.7: the circuit survives. (d) Conditional diversity of $sq2$ falls only without real data.

over-represented among the top 1% most influential training examples. Accumulating pools lowers the gain to 0.62, and a 5k pool raises it above 1 (1.03 vs 0.91 at 20k). A label-noise pulse decays at the predicted rate with real data (half-life 2.2 vs 2.1 generations at λ=0.25; 0.97 vs 0.93 at 0.5) but far more slowly without it (33 vs 14).

Table 1: Pre-registered temperature tests (confirmatory seeds 0–4). Only the qualitative prediction (test 5) held.

Test	Result	Observed ($T = 0.7, 1, 1.3, 1.6$)
1. $a(T)$ rises with T	fail	$a = -0.04, 0.93, 1.14, 1.04$
2. Runs inside 95% band	fail	3 of 12 cells
3. Contracting cells slow	fail	2 of 3 cells at T=0.7 (gain ≈ 0)
4. Runaway iff $a(1-\lambda) \geq 1$	fail	runaway only at λ=0; no seed in the 6 cells with $\lambda > 0$
5. Higher T: more off-context mass, less homogenisation	pass	mass 0.011, 0.079, 0.77, 0.82; entropy 0.40, 0.78, 0.91, 0.93

The pre-registered prediction failed. Higher temperature does raise hallucination and lowers homogenisation (Table 1, test 5). But the predicted critical fractions ($\lambda^* \approx 0.31$ at T=1.3, 0.52 at T=1.6) do not exist. Mass exceeded 0.5 in every seed at λ=0, yet none of the six cells predicted to run away with 20–50% real data did, in any seed.

Discussion

In the prompt-real regime, recursive training leaves the induction circuit intact: every self-generated query is still answered by copying from a real context, which re-teaches the same computation. Collapse concentrates in the free choice instead, mainly as off-context hallucination, and its growth is well described by a linear inheritance of errors through the data, which scrubbing and attribution identify causally. The failed pre-registration bounds that account: at high temperature, real data damps errors far more than a constant gain allows, so the law describes collapse locally but does not locate tipping points (its single-seed calibration also underestimated the gain at T=1: 0.86 vs 0.95). The circuit result needs real prompts: trained on its own full sequences without real data, the model's accuracy falls to 0.88 and its causal layer-1 effect from 0.83 to 0.71 by generation 9. Further limits are the 35k-parameter model, 16 symbols, 5–10 seeds and at most 20 generations. Two lessons carry over: accuracy can hide eroding calibration, and a modest share of real data, or accumulating data, sharply limits what a model inherits.

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Contribution Statements (Not in Page Limit)

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